

4.1 Answer each of the following:

- Suppose that a simple regression has quantities $N = 20$, $\sum y_i^2 = 7825.94$, $\bar{y} = 19.21$, and $SSR = 375.47$, find R^2 .
- Suppose that a simple regression has quantities $R^2 = 0.7911$, $SST = 725.94$, and $N = 20$, find $\hat{\sigma}^2$.
- Suppose that a simple regression has quantities $\sum (y_i - \bar{y})^2 = 631.63$ and $\sum \hat{e}_i^2 = 182.85$, find R^2 .

4.1

a.

$$SST = SSR + SSE$$

$$\sum (y_i - \bar{y})^2 = \sum (\hat{y}_i - \bar{y})^2 + \sum (y_i - \hat{y}_i)^2 \quad \bar{y} = \frac{\sum y_i}{N} \Rightarrow \sum y_i = N\bar{y}$$

$$\begin{aligned} \Rightarrow SST &= \sum (y_i - \bar{y})^2 = \sum (y_i^2 - 2y_i\bar{y} + \bar{y}^2) = \sum y_i^2 - 2\bar{y} \sum y_i + N\bar{y}^2 \\ &= \sum y_i^2 - 2N\bar{y}^2 + N\bar{y}^2 \\ &= \sum y_i^2 - N\bar{y}^2 = 7825.94 - 20 \times (19.21)^2 = 445.458 \end{aligned}$$

$$R^2 = \frac{SSR}{SST} \Rightarrow \frac{375.47}{445.458} = 0.84288 \#$$

b.

$$SST = SSR + SSE$$

$$\sum (y_i - \bar{y})^2 = \sum (\hat{y}_i - \bar{y})^2 + \sum (y_i - \hat{y}_i)^2$$

$$SSR = SST \times R^2 = 725.94 \times 0.7911 = 574.29$$

$$SSE \Rightarrow SST - SSR = 725.94 - 574.29 = 151.649$$

$$\hat{\sigma}^2 = \frac{SSE}{N-2} \Rightarrow \hat{\sigma}^2 = \frac{151.649}{20-2} = 8.425 \#$$

c.

$$SST = SSR + SSE$$

$$\sum (y_i - \bar{y})^2 = \sum (\hat{y}_i - \bar{y})^2 + \sum (y_i - \hat{y}_i)^2 \quad \sum \hat{e}_i^2$$

$$\Rightarrow SSR = SST - SSE$$

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST} = 1 - \frac{182.85}{631.63} = 0.7105 \#$$

4.3 We have five observations on x and y . They are $x_i = 3, 2, 1, -1, 0$ with corresponding y values $y_i = 4, 2, 3, 1, 0$. The fitted least squares line is $\hat{y}_i = 1.2 + 0.8x_i$, the sum of squared least squares residuals is $\sum_{i=1}^5 e_i^2 = 3.6$, $\sum_{i=1}^5 (x_i - \bar{x})^2 = 10$, and $\sum_{i=1}^5 (y_i - \bar{y})^2 = 10$. Carry out this exercise with a hand calculator. Compute

$$\bar{x} = 1$$

$$\hat{\sigma}^2 = \frac{SSE}{N-2} = 1.2$$

- the predicted value of y for $x_0 = 4$.
- the $se(f)$ corresponding to part (a).
- a 95% prediction interval for y given $x_0 = 4$.
- a 99% prediction interval for y given $x_0 = 4$.
- a 95% prediction interval for y given $x = \bar{x}$. Compare the width of this interval to the one computed in part (c).

4.3

a.

$$\hat{y}_i = 1.2 + 0.8x_i \Rightarrow \hat{y}_0 = 1.2 + 0.8 \times 4 = 4.4 \#$$

b.

$$\hat{\sigma}^2 = \frac{\sum e_i^2}{N-2} = \frac{3.6}{3} = 1.2$$

$$\hat{\text{var}}(f) = \hat{\sigma}^2 \left[1 + \frac{1}{N} + \frac{(x_0 - \bar{x})^2}{\sum (x_i - \bar{x})^2} \right] = 1.2 \times \left(1 + \frac{1}{5} + \frac{9}{10} \right) = 2.52$$

$$se(f) = \sqrt{\text{var}(f)} = 1.5875 \#$$

c.

$$95\% \text{ P.I.} \Rightarrow \hat{y}_0 \pm t_c se(f) \Rightarrow 4.4 \pm 3.1824 \times 1.5875 = [-0.652, 9.452] \#$$

d.

$$99\% \text{ P.I.} \Rightarrow \hat{y}_0 \pm t_c se(f) \Rightarrow 4.4 \pm 5.8409 \times 1.5875 = [-4.872, 13.672] \#$$

e.

$$x = \bar{x} = 1$$

$$\hat{y}_0 = 1.2 + 0.8 \times 1 = 2$$

$$\hat{\text{var}}(f) = \hat{\sigma}^2 \left[1 + \frac{1}{N} + \frac{(x_0 - \bar{x})^2}{\sum (x_i - \bar{x})^2} \right] = 1.2 \times \left(1 + \frac{1}{5} + 0 \right) = 1.44$$

$$se(f) = \sqrt{\text{var}(f)} = 1.2$$

$$95\% \text{ P.I.} \Rightarrow \hat{y}_0 \pm t_c se(f) \Rightarrow 2 \pm 3.1824 \times 1.2 = [-1.819, 5.819] \#$$

width of interval for part c = 10.10

width of interval for part e = 7.64

★ The interval in part (e) is narrower. The standard error of the forecast is minimized when the prediction point (x_0) is exactly the sample mean ($\bar{x}$). As the prediction point moves further away from the mean, the uncertainty increases, resulting in a wider prediction interval. #

4.25

a.

- $\ln(PRICE) = \beta_1 + \beta_2 SQFT + e$
- $b_2 = 0.036$ → A one-unit (square foot) increase in sqft leads to approximately 3.6% increase in house price.
- Slope: The estimated slope evaluated at the mean is $\beta_2 \times y = 9.0197$.
- Elasticity: The estimated elasticity evaluated at the mean is $\beta_2 \times x = 0.9834$.

b.

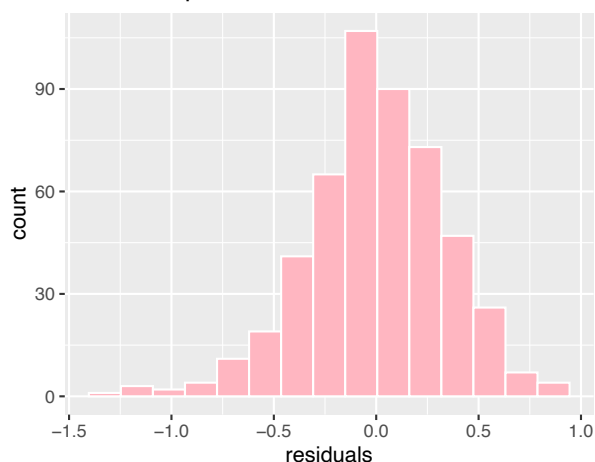
- $\ln(PRICE) = \alpha_1 + \alpha_2 SQFT + e$
- $a_2 = 1.0248$ → This represents the constant elasticity. A 1% increase in sqft leads to a 1.0248% increase in price. Since $\beta_2 > 1$, price is an increasing function of sqft at an increasing rate.
- Slope: The estimated slope evaluated at the mean is $\beta_2 \times y/x = 9.399$.
- Elasticity: The estimated elasticity evaluated at the mean is $\beta_2 = 1.0248$.

c.

- The R^2 for the linear model is 0.6413, the log-linear model is 0.6622, and the log-log model is 0.6445.
- Note that we cannot directly compare R^2 of the linear model with the other two because their dependent variables are in different scales. However, between log-linear and log-log models, the log-linear model yields a slightly higher R^2 , indicating a marginally better fit for the data in the logarithmic scale.

d.

Residuals plots of a



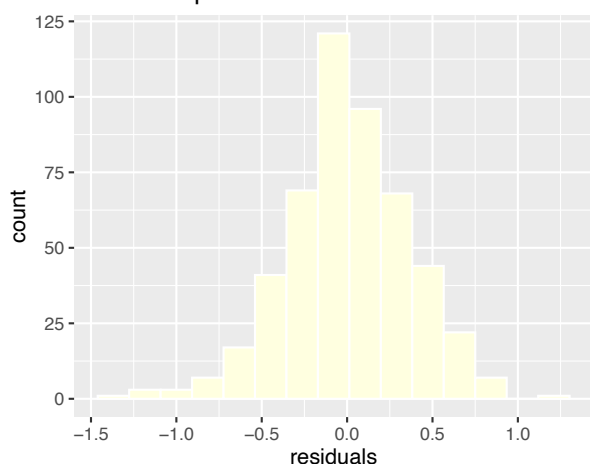
```
> jarque.bera.test(ehat_a)
```

Jarque Bera Test

data: ehat_a
X-squared = 26.679, df = 2, p-value = 1.61e-06

$p\text{-value} < 0.05 \Rightarrow \text{reject } H_0$
 $\Rightarrow$ The data is not normally distributed.

Residuals plots of b



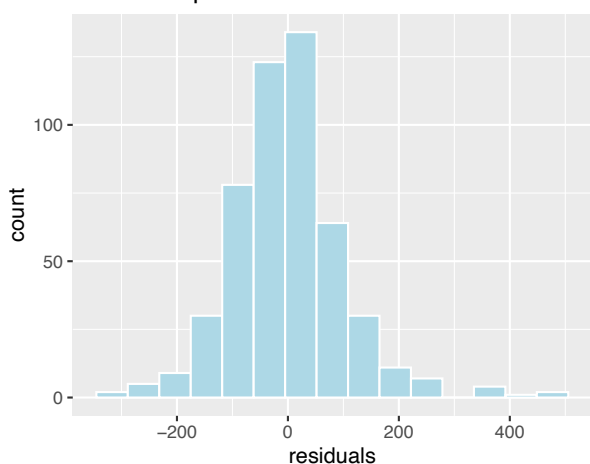
```
> jarque.bera.test(ehat_b)
```

Jarque Bera Test

data: ehat_b
X-squared = 14.279, df = 2, p-value = 0.0007933

$p\text{-value} < 0.05 \Rightarrow \text{reject } H_0$
 $\Rightarrow$ The data is not normally distributed.

Residuals plots of c



```
> jarque.bera.test(ehat_c)
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Jarque Bera Test

data: ehat_c
X-squared = 221.11, df = 2, p-value < 2.2e-16

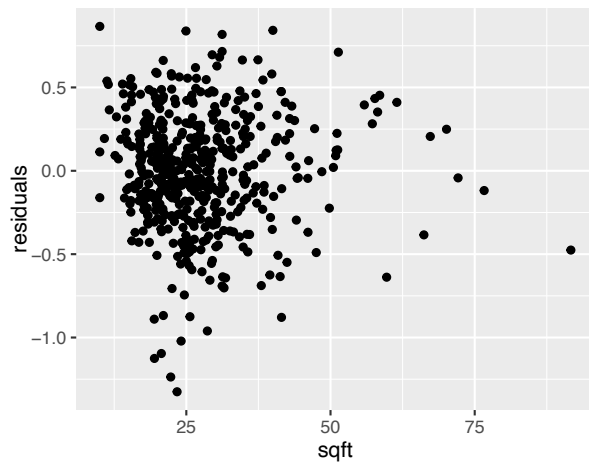
$p\text{-value} < 0.05 \Rightarrow \text{reject } H_0$
 $\Rightarrow$ The data is not normally distributed.

- Based on the histograms, the residuals in the log-linear (model a) and log-log (model b) models appear relatively symmetric and are generally compatible with the normality assumption. However, in the linear model (model c), the residual distribution is slightly right

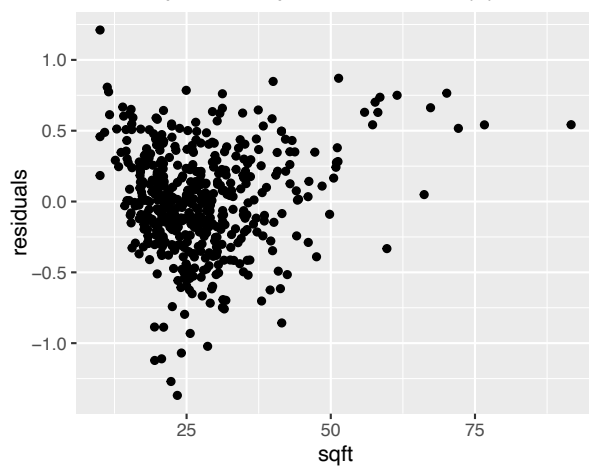
skewed, which clearly violates the assumption of normally distributed errors.

e.

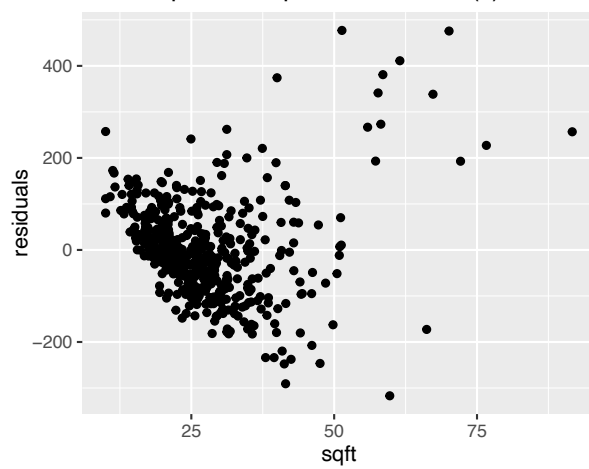
Scatter plots of sqft and residuals (a)



Scatter plots of sqft and residuals (b)



Scatter plots of sqft and residuals (c)



- None of the residual scatter plots display a perfectly random spread. Specifically, model c (linear model) exhibits a clear “funnel shape”, indicating severe heteroskedasticity (the variance of residuals increase as sqft increases). Applying the log transformation in models a and b mitigates this issue significantly.

f.

- For a house with 2700 square feet, the predicted values are:
 - Model a (log-linear): 214.2336
 - Model b (log-log): 227.5386
 - Model c (linear): 246.4557

g.

- Model a: The 95% prediction interval is [109.7685, 418.1167].
- Model b: The 95% prediction interval is [111.1406, 465.8406].
- Model c: The 95% prediction interval is [44.2772, 448.6341].

h.

- Conclusion:
 - Based on the analyses above, I will choose the log-linear model (model a). While it is invalid to compare its R^2 directly with the linear model, it outperforms the log-log model in explanatory power (R^2). More importantly, the log transformation effectively addresses the severe right-skewness and heteroskedasticity observed in the linear model's residuals, making the statistical inferences in the log-linear model much more reliable.